



Notes for submissions

- Please pack your solution into a **single zip-archive**.
- If I cannot execute (e.g. Python/AWK) or compile (e.g. C, C++) your code, you will get 0 points. If there is something non-obvious necessary for compiling or running the code, please add a README file.

Exercise 1: Visualization of Monte Carlo moves

(4 points)

Unpack the tar ball *pivot.tar.gz* and compile the program with “*make mc*”. Go to the folder *TEST* and run the program with “*./P1000 eingabe*”. This is a short MC simulation run with one polymer chain using local and pivot moves.

Change the file “*eingabe*” to write out every step into a trajectory file (*out* = 1, *trajectory* = 1). Hand in trajectories for all parts of this exercise.

- Alter the file “*eingabe*” in a way that only local moves are done. Simulate a short run, visualize your results with VMD. Hand in a picture of the result created by VMD. To do this you can use *File/Render..* in VMD.
- Repeat exercise (a) but use only reptation moves.
- Repeat exercise (a) but use only pivot moves.
- What do you observe? Which move is the most effective to generate uncorrelated configurations? Why are some frames of your trajectory identical to the previous?
- Change the temperature in “*conf_out*” to $T = 1$ and use a combination of the three different moves from above. What happens?

Exercise 2: Scaling behavior of polymer chains

(4 points)

There is a scaling law for the end-to-end radius (as well as the gyration radius): $R_{ee}^2 \sim N^{2\nu}$. We want to confirm this with computer simulations at $T = 4.98$.

- Make a longer simulation run for values of $N = 200, 400, 600, 800, 1000$. To make sure that the polymer is equilibrated, do at least 100,000 steps for $N = 1000$ and an appropriate amount of steps for the other lengths.
Warning: If you go from lower values of N to higher, remember to refresh your starting file “*conf_out*” by copying “*conf_start*” to “*conf_out*” every time!
- Calculate the quadratic end-to-end radius R_{ee}^2 values by averaging over the second column in “*analyse*”, e.g. using awk or python.
- Plot $R_{ee}^2(N)$ and determine ν .

Exercise 3: Random walk on a lattice

(8 points)

For polymers without volume exclusion there is a similar scaling law which can be calculated easily:

$$\begin{aligned}
 R_{ee}^2 &= \langle (\vec{r}_1 + \vec{r}_2 + \dots + \vec{r}_N) \cdot (\vec{r}_1 + \vec{r}_2 + \dots + \vec{r}_N) \rangle \\
 &= \langle \vec{r}_1 \cdot \vec{r}_1 + \vec{r}_1 \cdot \vec{r}_2 + \dots + \vec{r}_1 \cdot \vec{r}_N + \dots + \vec{r}_N \cdot \vec{r}_N \rangle \\
 &\stackrel{1}{=} \sum_{n=1, m=1}^N \vec{r}_n \cdot \vec{r}_m \delta_{nm} \stackrel{2}{=} l^2 N
 \end{aligned} \tag{1}$$

1: The average scalar product of two random vectors with no preferred direction is zero.

2: l is the average bond length of this random walk.

(a) Write your own program to simulate a random walk on a lattice.

Note: Monomers do not have a volume, therefore, they can be placed one after the other and no Metropolis Monte Carlo steps are needed. (This does not work for monomers with volume exclusion as in exercise 2 and 3 as the whole chain would have to be discarded if there is a single overlap.)

(b) Simulate a random walk with $N = 200, 400, 600, 800, 1000$ and show that $\nu = 0.5$.

Exercise 4: Maxwell-Boltzmann distribution / thermal de Broglie wavelength

(4 points)

(a) Plot the velocity distribution for SiO_2 at its glass transition temperature of $T_c = 1450K$.

(b) Show that the average velocity of a particle moving according to the Maxwell-Boltzmann distribution is given by

$$\langle v \rangle = \sqrt{\frac{8k_B T}{m\pi}} \tag{2}$$

(c) In quantum mechanics a wavelength for moving particles can be defined according to the wave-particle duality. Comparing such de Broglie wavelengths to typical length scales of the system (e.g. bond lengths) allows a rough estimate whether or not quantum mechanical effects are relevant when observing the system.

Calculate the de Broglie wavelength according to the average Maxwell-Boltzmann velocity

$$\lambda = \frac{h}{p} \tag{3}$$

and express it as a multiple of the thermal de Broglie wavelength

$$\lambda_T = \frac{h}{\sqrt{2\pi m k_B T}}. \tag{4}$$

(d) Calculate the thermal de Broglie wavelength for SiO_2 at its glass transition temperature of $T_c = 1450K$. Compare this value with the bond length of a $Si - O$ bond ($\sim 1.6 \text{ \AA}$). Is a quantum mechanical treatment of the system necessary?